Li 2017 CRC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9/10 samples (248 malignant cells) intermix on one continuous, elongated manifold (radius-graph: 0 detached islands; kNN: 1 component); whole manifold enters the trajectory.

Gate 2 — module geography → GO (non-branching)

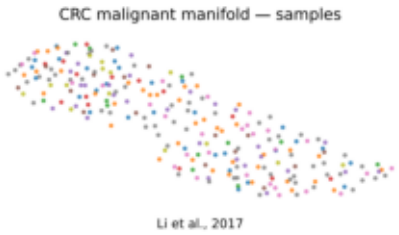
Gast/pSom co-vary (r=+0.89) and decline along pseudotime (ρ −0.82 / −0.37); Crest anti-correlated (r=−0.96 / −0.97), rising to a terminal readout (ρ +0.56; crest-high decile pt=1.00).

Gate 3 — cross-repression → NA

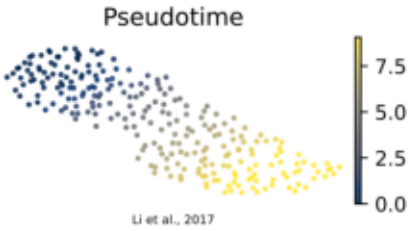
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

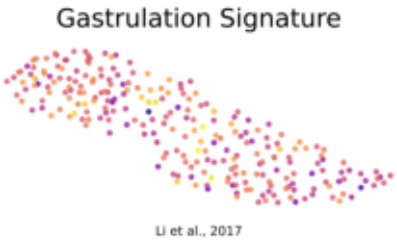
Manifold — samples



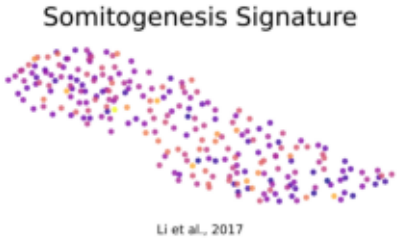
Pseudotime



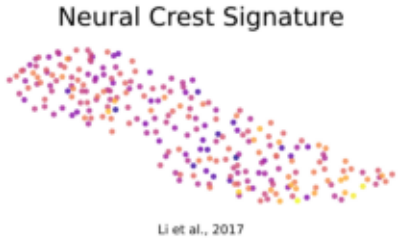
Gastrulation



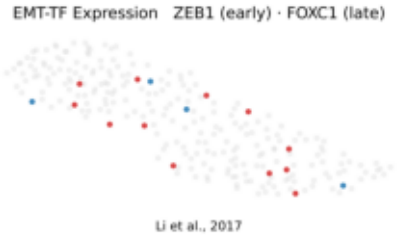
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Colorectal carcinoma (Li et al., 2017), SmartSeq2 — 18 samples / 590 cells (master scope); 254 malignant loaded; 248 across 9 patients (≥8 cells) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory. Non-branching Linear-pSom/Crest.

Verdict.

Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Li's colorectal malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Cell cycle is orthogonal to the axis (pseudotime↔cycling ρ=−0.01), so the whole manifold is kept. EMT-TFs ZEB1 (early/root) → FOXC1 (late/terminus); end-separation weak (Δpt=0.14) at this SmartSeq2 depth (248 cells).

Parameter choices

Min cells/sample	≥8 (keeps 9/10; smooth counts)	**Integration**	Harmony θ=5; PCA(HVG); spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=15, λ=0.3, μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**HVG**	seurat 2000 (pagoda2 degenerate at n=248)
EMT-TFs	ZEB1 (early/root) → FOXC1 (late/terminus); Δpt=0.14		